SOFTWARE PROJECT LAB 1

ARGUS

Advanced Real-Time Exam Supervision Engine

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Project Description

"Because cheating shouldn't be easier than studying."

Argus is a high-performance network monitor engineered to eliminate digital examination loopholes with absolute zero network latency.

- **Native OS Integration:** Operates directly at the packet layer using raw sockets.
- **Advanced Inspection:** Bypasses standard application-layer limitations and simple DNS filters.
- **Threat Neutralization:** Defeats hidden encrypted VPN tunnels and proxy chains.
- **Anti-Tampering:** Systematically prevents local OS clock manipulation.

Motivation

The Problem

Basic DNS filtering is highly vulnerable. Students easily bypass these measures using proxy chains and standard circumvention tools during exams.

The Vulnerability

Undetected encrypted VPN tunnels and local OS clock tampering entirely compromise the integrity of digital examinations across the network.

The Solution

A robust, low-level Deep Packet Inspection (DPI) engine designed to ensure fundamentally fair and secure environments for honest students.

System Workflow

1. Capture

Intercepts traffic silently at native OS layer via raw sockets.

2. Parse

Strips Ethernet, IP, and TCP headers mathematically in-memory.

3. Extract

Parses TLS Client Hello handshake for Server Name Indication.

4. Enforce

Evaluates parsed hostname via an Aho-Corasick Automaton.

5. Drop

Proactively drops and logs unauthorized data packets.

Core System Features

- **Deep Packet Inspection:** Bypasses DNS filtering by extracting Server Name Indication from TLS handshakes prior to encryption.
- **Hybrid Cryptography:** Secures telemetry tunnel using a robust mix of asymmetric (RSA) and symmetric (RC4/XOR) encryption.
- **Precision Anti-Tamper:** Syncs clients with a trusted server time, actively defeating local OS clock manipulation.
- **VPN Detection:** Analyzes payload randomness using Shannon Entropy to identify tunnels without data decryption.
- **Multiplexed Dashboard:** Single-threaded architecture asynchronously monitoring students via an ImGui/OpenGL interface.
- **Data Compression:** Huffman Coding shrinks violation logs to fit strictly within the 512-byte binary protocol limit.

The Algorithmic Engine

Rivest–Shamir–Adleman (RSA)

A widely used asymmetric (public-key) cryptographic method developed in 1977, essential for securing online data transmission

Shannon Entropy

Identifies VPNs via payload randomness. Encrypted blocks yield entropy approaching 8.0.

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_2 (P(x_i))$$

Cristian's Algorithm

Synchronizes client/server time dynamically to calculate offsets and defeat local clock tampering.

$$Offset = \frac{T_{recv} - T_{sent}}{2}$$

Huffman Coding

Prefix-free variable-length encoding generated via a Min-Heap frequency tree to compress 512–byte payload limits securely without data loss.

Technology Stack

Core Engine

Written entirely in high-performance **C++17**. Utilizes strict packet-level network control via **libpcap** and native raw sockets.

Supervision UI

Renders a lightweight, responsive, and hardware-accelerated dashboard interface utilizing **Dear ImGui**, **GLFW**, and **OpenGL 3.2+**.

Version Control

Strict source code versioning, branch management, and collaborative integration managed via **Git** and hosted securely on **GitHub**.

Thank You

https://github.com/DevNexus09/Argus-ExamSupervisonSystem.git